

Literature status: $\text{Tame}(L^1(G)) = \text{Tame}(G)$ for locally compact groups

Run `fa_banach_001`; source arXiv:1710.01044

Source question

Michael Megrelishvili's paper *Tame functionals on Banach algebras* introduces tame functionals on Banach algebras and proves $\text{Tame}(\ell^1(G)) = \text{Tame}(G)$ for discrete groups. It then asks whether the same equality holds for nondiscrete locally compact groups.

we are going to investigate tame functionals in some future works. Among others we intend to deal with the following natural question.

Question 3.6. *Is it true that $\text{Tame}(L_1(G)) = \text{Tame}(G)$ for every (nondiscrete) locally compact group G ?*

Figure 1: Question 3.6 in arXiv:1710.01044, page 8.

The extracted question is:

Is $\text{Tame}(L_1(G)) = \text{Tame}(G)$ for every nondiscrete locally compact group G ?

Supporting literature status

Matan Komisarchik's paper *The Banach Algebra $L^1(G)$ and Tame Functionals* explicitly identifies this as Megrelishvili's question and answers it affirmatively. Its introduction states the main result as a general bornological equivalence and lists the tame equality as an immediate consequence.

(2) φ is *right $\mathcal{B}$ -small as a functional* if $\varphi \otimes B_{L^1(G)} \in \mathcal{B}$ (Definition 5.1). In this case we write $\varphi \in \text{small}_{a,r}(\mathcal{B}, \text{RUC}_b(G))$.

A detailed study of tame and Asplund subsets as norm-saturated, convex vector bornologies, can be found in [24].

Theorem 6.3. For every norm-saturated, convex vector bornology $\mathcal{B}$ on $\text{RUC}_b(G)$ we have:

$$\varphi \cdot G \in \mathcal{B} \iff \mathfrak{L}_\varphi(B_{\text{RUC}_b(G)}) = \varphi \otimes B_{\text{RUC}_b(G)} \in \mathcal{B}.$$

In particular:

$$\text{Tame}(L^1(G)) = \text{Tame}(G)$$

$$\text{Asp}(L^1(G)) = \text{Asp}(G)$$

$$\text{WAP}(L^1(G)) = \text{WAP}(G).$$

Figure 2: arXiv:2301.12298, page 2: the introduction restates the main theorem and includes $\text{Tame}(L^1(G)) = \text{Tame}(G)$.

The result is proved later as Theorem 6.3.

Idea of the proof identification

This packet does not claim an original proof. The identification is direct: Megrelishvili's Question 3.6 asks exactly for the equality $\text{Tame}(L_1(G)) = \text{Tame}(G)$ for nondiscrete locally compact groups,

- [2] M. Komisarchik, *The Banach Algebra $L^1(G)$ and Tame Functionals*, arXiv:2301.12298v6, 2025.